Econometrics 1 — Problem Set 3

Submission Deadline: Thursday, October 15th, 2026 at 1:00 PM | Instructor: Liam Wren-Lewis, TA: Elisa Alonso Herrero

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0.1 Overview & Group Work Division

- **Exercise 1 (Theoretical / Reading Critique):**
-

0.2 Computer Exercise (Empirical Analysis in R):

1 Exercise 1: Theoretical & Reading Questions

> **!** Classmates' Assignment
>
> This section contains Exercise 1 (Part A and Part B). Space has been allocated below for your classmates to fill in their solutions and write-ups.

1.1 Part A: Article Critique

Reading: *"Legal determinants of World Cup Success"* (by Mark West).

1.1.1 Question 1: Econometric Strategy & Gauss-Markov Critique

Formulate a critique related to the econometric strategy used in the paper. Think about which of the Gauss-Markov assumptions might be violated in the regressions and what that tells you about the interpretation of the results. Limit your critique to one page.

> **i** Answer — Exercise 1.A.1
>
> **Violated Gauss-Markov Assumptions & Implications:**
>
> - **Zero Conditional Mean Assumption $E(u|X) = 0$ (Omitted Variable Bias):**
> - **Homoskedasticity / Uncorrelated Errors:**
> - **Interpretation of Coefficients:**

1.1.2 Question 2: Reliability of Estimations

The 4 different estimations in this paper seem to lead to quite different results. If you wanted to conclude anything related to the determinants of World Cup soccer success, would you rely on any of these estimations? If so, on which one?

Answer — Exercise 1.A.2

Assessment:

1.2 Part B: True or False

Missing observations can lead to both biased estimates and imprecise estimates. T/F? Explain.

Answer — Exercise 1.B

Statement: [True / False]
Explanation:

- **Biased Estimates:**
- **Imprecise Estimates:**

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2 Computer Exercise: Empirical Analysis in R

Your Assignment

This section contains the 7 empirical questions using the dataset `nichh.dta`. Code chunks are prepared below for execution in VS Code / R.

2.1 Data Setup & Model Estimation

Dataset: `nichh.dta` (subset of 1998 Nicaragua LSMS dataset, 582 households).

Task: Estimate a log-linear model that allows understanding the *ceteris paribus* relationships between total household consumption (in logs) and household head's age, head's education, male and female labor endowments, land endowments, and distances to the health clinic and school.

2.2 Question (1): Education Significance

What is the minimum significance level at which one can reject the hypothesis that the education level of the household head does not affect total household consumption?

i Answer — Question (1)

As shown in Table 1, the coefficient of the household head's education is statistically significant at the 1% significant level, with a p-value of $< 2e-16$ (almost zero). This means the minimum significance level at which we can reject the null hypothesis that the education level of the household head does not affect total household consumption is $< 2e-16$.

2.3 Question (2): Male vs. Female Labor Returns

Do your results allow you to conclude the returns to male adults in the household are larger than the returns to female adults? State the null hypothesis and the alternative hypothesis you are testing, and the significance level you are considering. Do the test and discuss. Does your answer differ depending on which significance level you consider?

i Answer — Question (2)

Recalling the model functional form:

$$\log(\text{ConsTotal}) = \beta_0 + \beta_1 \text{Adult}_{Male} + \beta_2 \text{Adult}_{Female} + \text{CZ} + u$$

Where Z represents the vector of control variables
and C is the vector of their coefficients.

Using Equation 2.3 the null and alternative hypotheses are as follows:

Table 1: Log-Linear model summary. Dependent variable: log(total household consumption).

	Base	No Age	Land Area <= 100
Intercept	9.178 ***	9.470 ***	9.181 ***
	(0.108)	(0.076)	(0.108)
Number of male adults	0.115 ***	0.126 ***	0.114 ***
	(0.025)	(0.025)	(0.025)
Number of female adults	0.195 ***	0.209 ***	0.193 ***
	(0.025)	(0.025)	(0.025)
Age of household head (years)	0.006 ***		0.006 ***
	(0.002)		(0.002)
Education of household head (years)	0.066 ***	0.060 ***	0.065 ***
	(0.005)	(0.005)	(0.005)
Land owned (ha)	0.004 ***	0.004 ***	0.010 **
	(0.001)	(0.001)	(0.004)
Distance to health centre (km)	-0.011	-0.010	-0.010
	(0.010)	(0.011)	(0.011)
Distance to school (km)	-0.020 *	-0.023 **	-0.020 *
	(0.011)	(0.011)	(0.011)
N	575	575	569
R2	0.354	0.338	0.336

Significance: *** p < 0.01; ** p < 0.05; * p < 0.1

$$H_0 : \beta_1 \leq \beta_2$$
$$H_1 : \beta_1 > \beta_2$$

```
   adultm
-2.090923
```

Running a one sided t-test to verify that $\beta_1 - \beta_2 > 0$, the results show a t-statistic of -2.090923 which t is less than any critical value for the significance levels of 1%, 5%, or 10%. In fact, for one sided tests any t-statistic less than zero will fail to reject the null hypothesis because its p-value will be greater than the significance level. Therefore, we can not conclude that returns to male adults in the household are larger than the returns to female adults regardless of the significance level considered.

2.4 Question (3): Joint Significance of Distance Variables

Test whether the 2 distance variables are jointly significant. Calculate the test by hand (using R). What can you conclude about the role of isolation on household welfare?

i Answer — Question (3)

Let's estimate the restricted and unrestricted models and calculate the F-statistic by hand:

```
[1] 2.593946
```

```
[1] 2.311961
```

Using the standard formula for the F-statistic:

$$\begin{aligned} F &= \frac{(RSS_R - RSS_U)/q}{RSS_U/(n-k-1)} \\ &= \frac{(192.4492 - 190.7043)/2}{190.7043/(575-7-1)} = 2.59 \end{aligned}$$

Since the critical value for the F-distribution with 2 restrictions and 567 degrees of freedom is 3.01, we fail to reject the null hypothesis that the two distance variables are jointly insignificant. However, we can reject the null hypothesis at a 10% significance level, as the critical value for that level is 2.31. This suggest that isolation (measured by distance to health clinic and school) has a weak but significant effect on household welfare, and that the two distance variables are jointly significant at a 10% significance level.

2.5 Question (4): Correlation Between Education and Age

Analyze the correlation between households' education and age. What does this correlation imply for your estimates? If you are estimating this model to obtain an estimate of the returns to household's heads education on household welfare, should you keep household age in the model?

i Answer — Question (4)

The correlation between household's education and age is -0.31 and should be included as control in the functional form to increase precision for the following reasons.

- Since age affects education and consumption and not the education affects age because by definition it is not possible, age is a *good control* to include. Adding it should increase the precision of the estimators.
- Omitting age would downward bias the estimated effect of education on consumption. Given the correlation of -0.31 and an effect of age on consumption of 0.006, omitting age would downward bias our estimate by $corr(education, age) \times \beta_{age} = 0.31 \cdot 0.006 = 0.00186$.

2.6 Question (5): Testing Coefficient equal to 0.06

Test whether the coefficient of education is different from 0.06.

Answer — Question (5)

The null and alternative hypotheses are:

$$H_0 : \beta_{headeduc} = 0.06$$
$$H_1 : \beta_{headeduc} \neq 0.06$$

```
[1] 1.171443
```

```
[1] 0.07535767
```

```
[1] 0.05613264
```

Estimating the t-statistic using $t = \frac{\hat{\beta} - 0.06}{SE(\hat{\beta})}$, we obtain a t-statistic of 1.171443, which is less than the critical value of 1.96 for a two-sided test at the 5% significance level. Therefore, we fail to reject the null hypothesis that the coefficient of education is equal to 0.06. In other words, we do not have enough evidence to conclude that the coefficient of education is different from 0.06 at the 5% significance level.

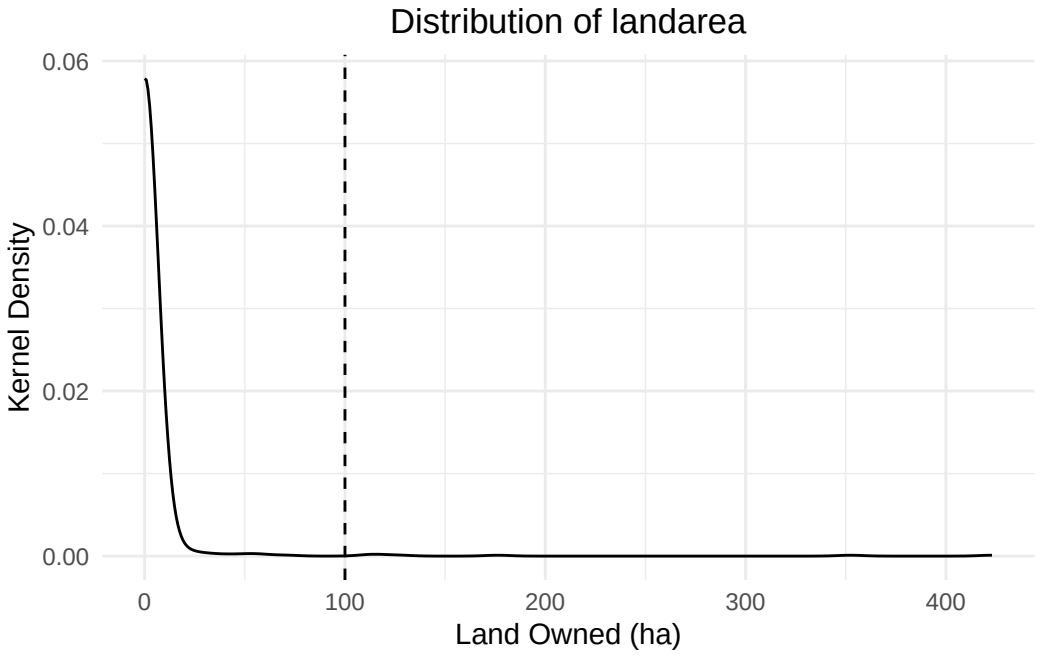
2.7 Question (6): Outliers in Land Area Variable

Look at the distribution of the landarea variable. Then re-estimate the model excluding households with more than 100 ha. Analyze the change of the coefficient of the land area variable compared to the previous regression. Explain.

Answer — Question (6)

Looking at the distribution of the landarea variable in Figure 1, we can see that there are a few households with land area greater than 100 ha that create a long right tail in the distribution. Most of the hoseholds (87.5%) have a land area equal to zero so the distribution is heavily skewed. In theory removing outliers should increase the estimate of the land area coefficient the regression line will have a stepper slope.
As shown in Table 1, the coefficient of land area in the model with all households is 0.004 with a standard error of 0.001, while the coefficient of land area in the model excluding households with more than 100 ha is 0.010 with a standard error of 0.004 no longer significant at the 5% significance level but at 10% significance level. The rest of the coefficients remain unchanged indicating the base model is robust to the exclusion of outliers.

Figure 1: Distribution of Land Area (ha)



2.8 Question (7): Omitted Variable Thought Experiment

Give an example of an additional variable you might want to include in the regression, but for which you don't have data, and indicate the expected effect of that variable, as well as any

changes you would anticipate to occur in the point estimates of the other variables.

Answer — Question (7)

An additional variable that could be included in the regression is *presence of chronic deseases* in the hosehold head. Since deseases can affect the access to education and the ability to work affecting income and consumption, we would expect a negative effect of this variable on household consumption. It is also a good control because it affects education and consumption.If this variable is omitted from the regression, we would expect the coefficient of the education variable to be biased upwards, as households without chronic diseases may have higher levels of education and thus higher consumption.